



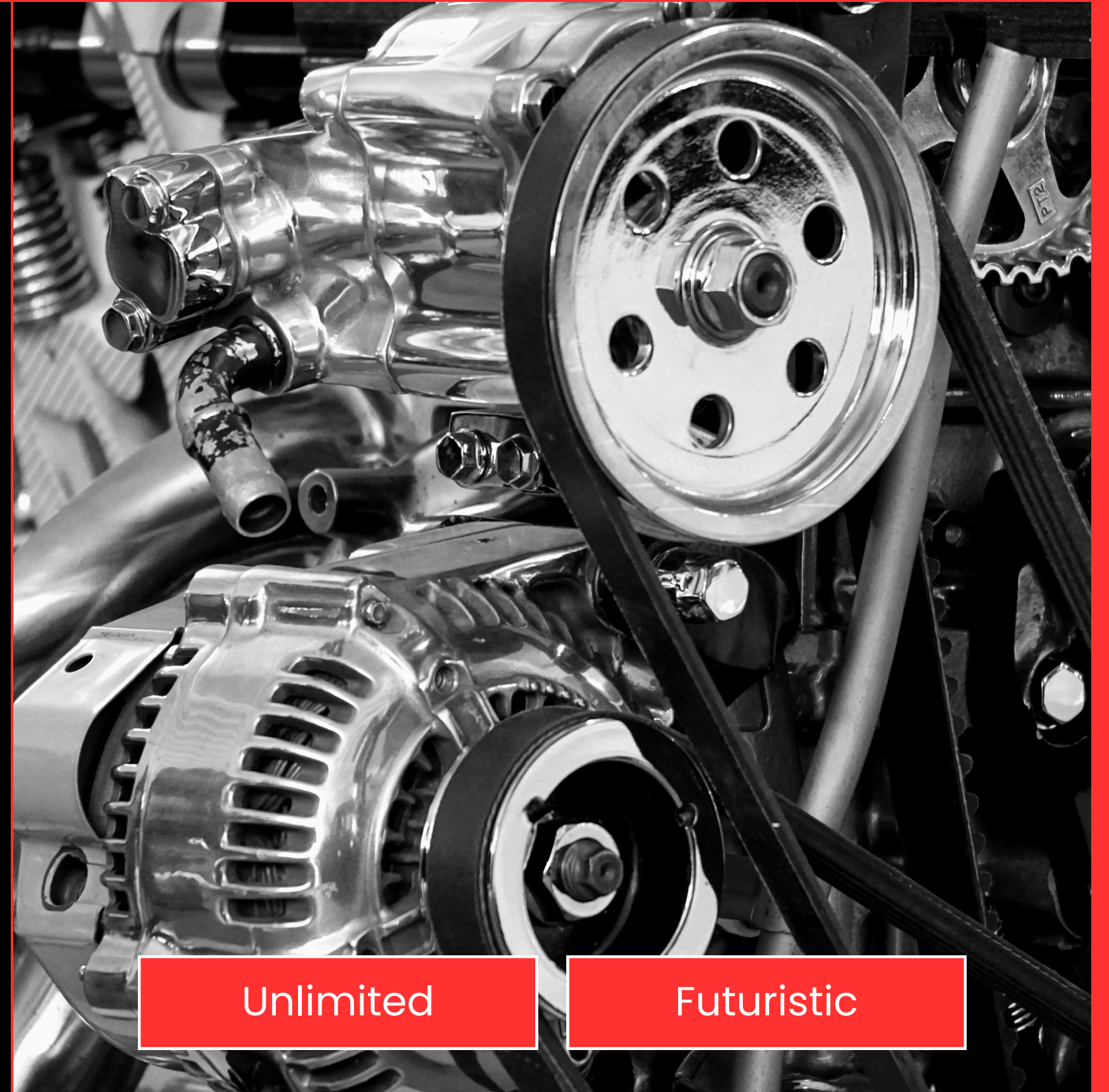
Engineering



AI Engineering / Mechanical
Prompt Engineering

Unlimited

Futuristic





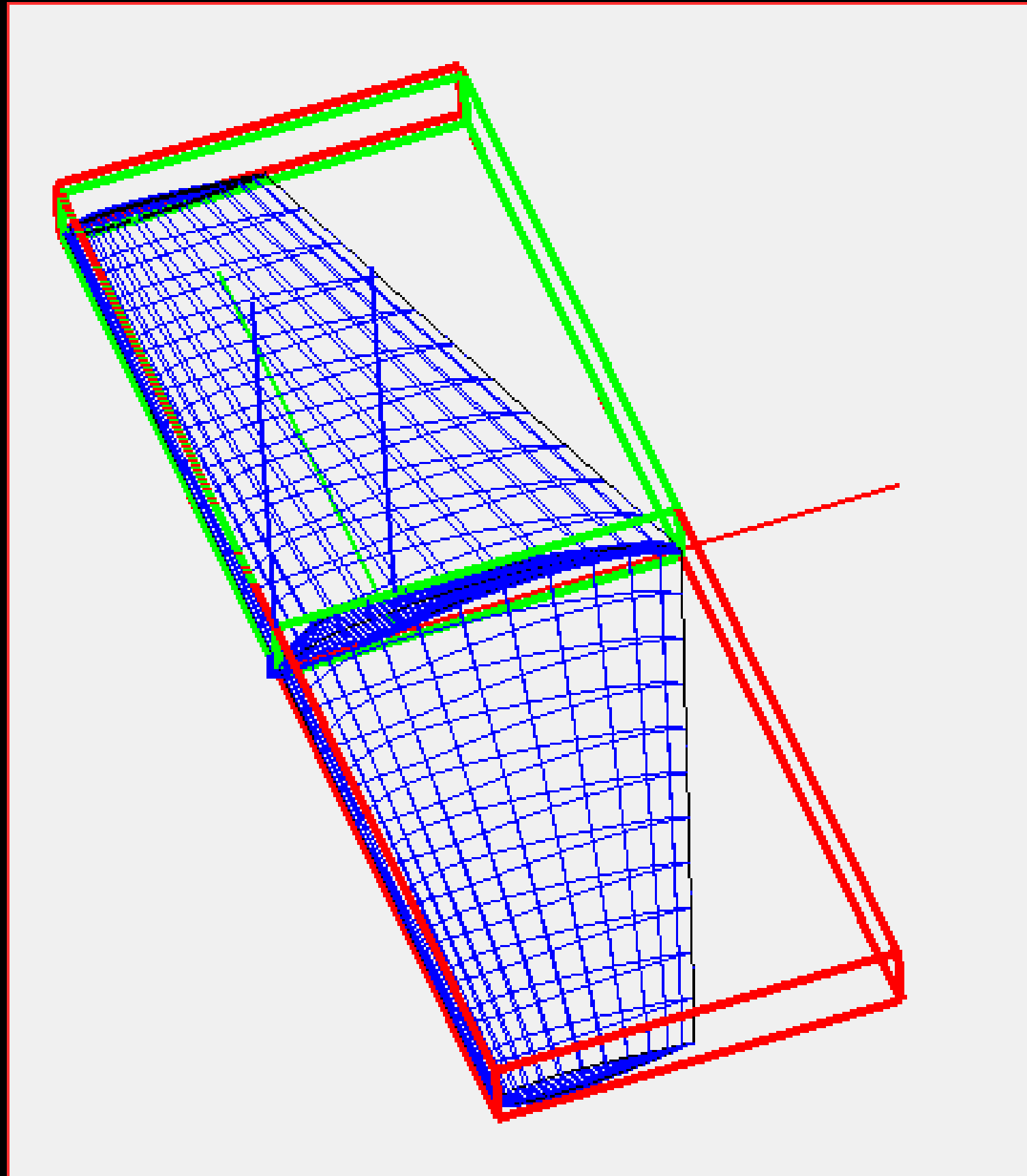
About me

Turning Ideas Into Reality

Mechanical Engineer & AI Prompt Engineer

Bridging classical mechanical design, thermodynamics, and materials science with advanced LLM prompting and script generation. Focused on leveraging AI to accelerate CAD workflows, FEA setups, and thermal analysis without sacrificing physical rigor.





FEA & Mechanical



Fields That Drive Progress

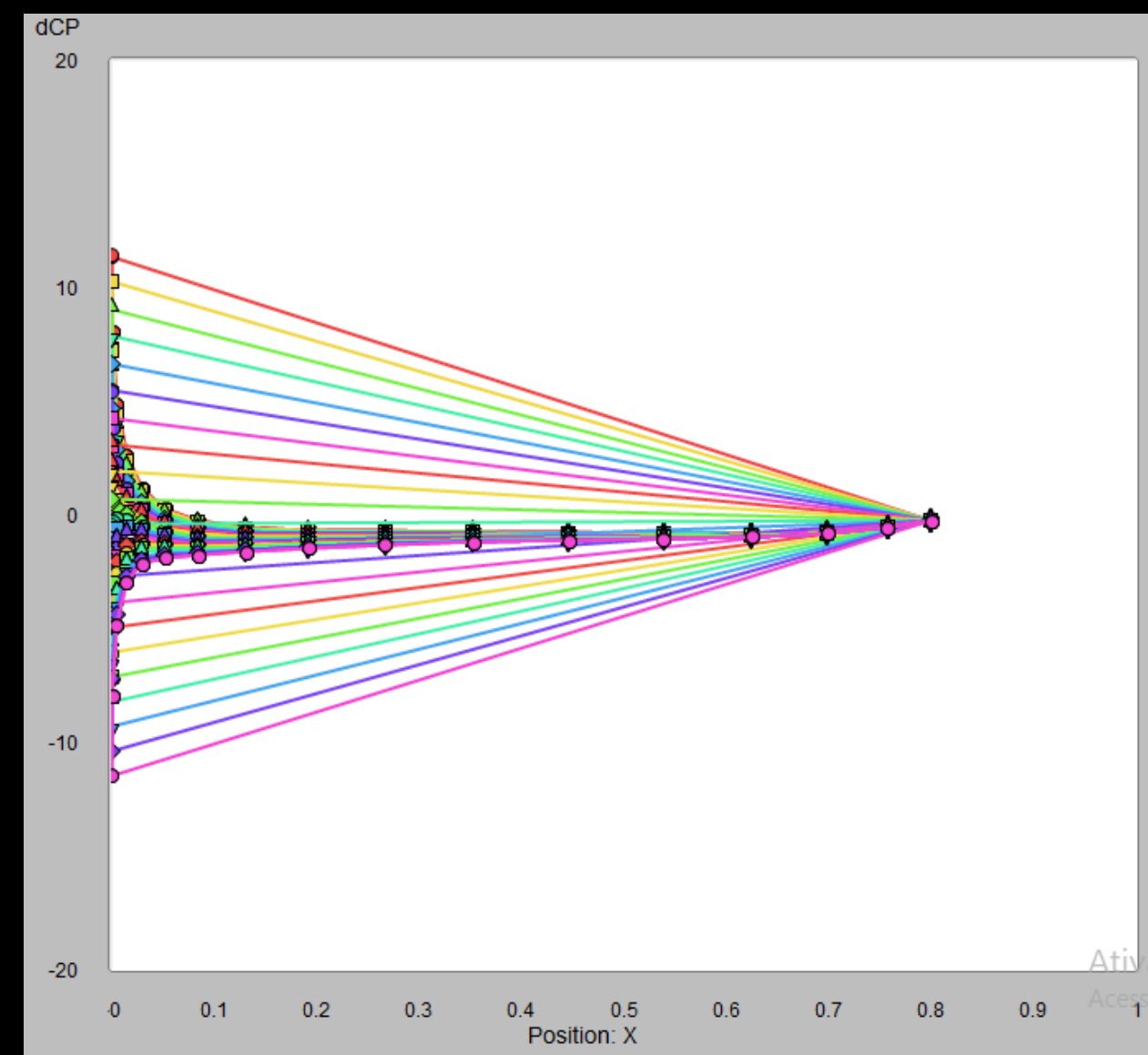
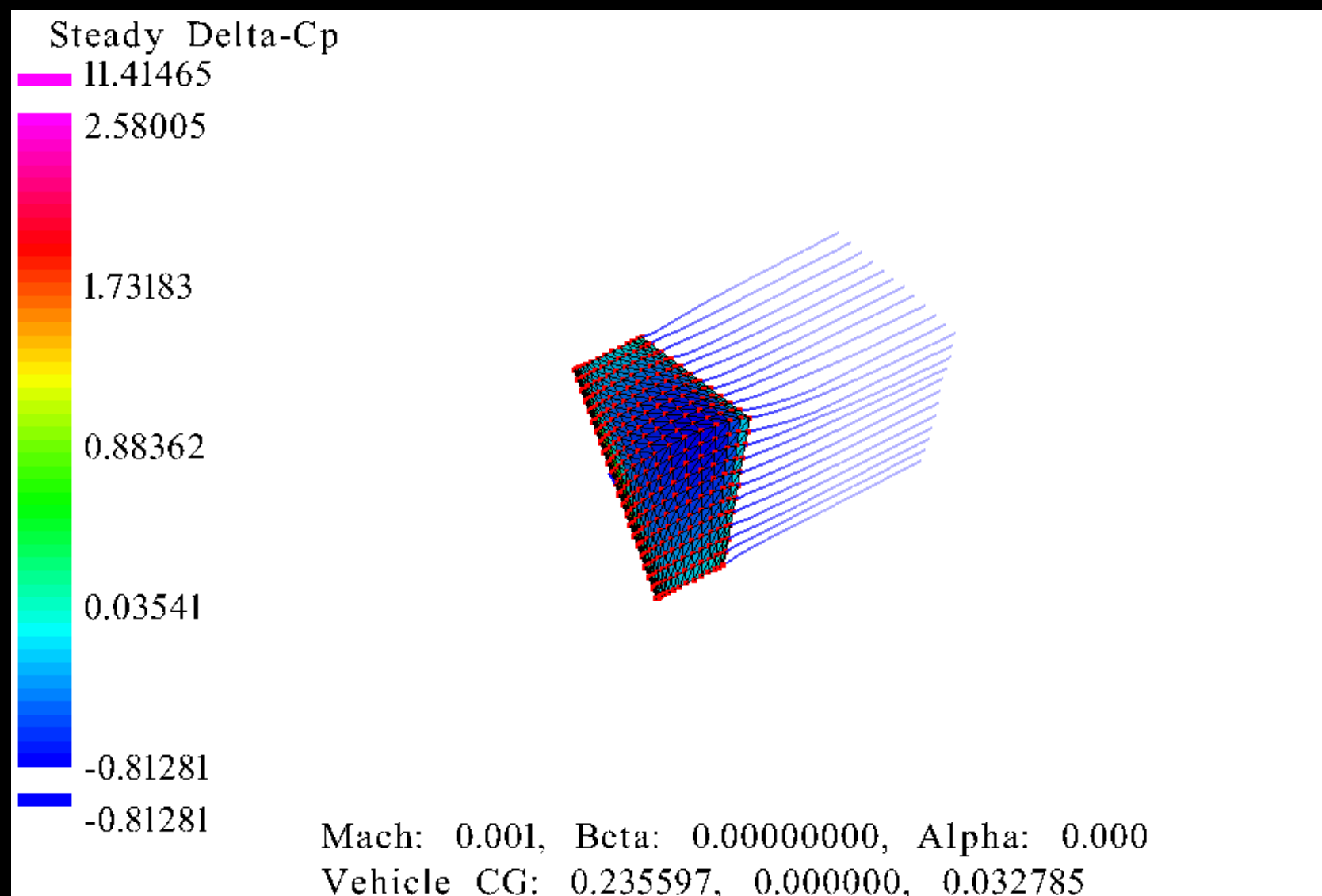
I selected a real capture of a structural FEA simulation from an wing of an aircraft model I built. The screenshot showcases the finite element mesh and von Mises stress distribution. Using an authentic engineering capture instead of an AI-generated 3D render proves real modeling experience and technical accuracy.



The Process

From Concept to Creation

Heat transfer is governed by strict physical laws (conservation of energy, fluid boundary layers, and convective coefficients). Generative AI renders of "thermal imaging" create colorful, visually striking heat maps that violate thermodynamics and lack physical validity.





AI & Prompting

Wonders Large and Small

To automate the geometry setup in OpenVSP and run batch VSPAERO numerical simulations for my aerodynamics study, I designed a structured, physics-constrained prompt architecture:

[ROLE / SYSTEM]

You are a Senior Aeronautical Engineer and Computational Fluid Dynamics (CFD) specialist expert in parametric geometry modeling (OpenVSP / Vehicle Sketch Pad), Vortex Lattice Method (VLM) simulations (VSPAERO), and aerodynamic validation (XFLR5 / Panel Methods).

[CONTEXT]

I am conducting an aerodynamic analysis and numerical simulation of an RC aircraft wing (Safer AeroDesign scale).

Key parameters of the wing model:

- Airfoil: Selig-1223RTL (high-lift profile)
- Wing Geometry: Trapezoidal, Span = 2.4 m, Root Chord = 0.6 m, Tip Chord = 0.3 m (Aspect Ratio = 4.0, Area = 1.44 m²)
- Tip Treatment: Round Cap, SkewBoth configuration, with LE/TE clustering for mesh refinement
- Mass / CG: Total mass = 1.0 kg, CG located at X = 0.236 m from the wing root
- Flight Conditions: Subsonic flow, Mach ~ 0.001, Reynolds Number (Re) ranged between 10⁴ and 10⁶, Angle of Attack (Alpha) sweep from 0° to 10° (21 evaluation points).

[TASK]

1. Generate an automated Python script (using OpenVSP API / PyVSP or custom script) to programmatically build this wing geometry and set up the VSPAERO simulation case with the specified boundary conditions.
2. Provide a structured framework for cross-validating the aerodynamic coefficients (CL vs. Alpha, CDt vs. CL, and CMY vs. Alpha) against XFLR5 batch simulation outputs.
3. Explain the physical mechanism behind the stall behavior observed around Alpha = 6.5° to 10° for this specific Selig-1223RTL profile under low Reynolds numbers.

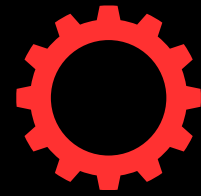
[CONSTRAINTS & TECHNICAL RIGOR]

- Ensure the Python script correctly handles OpenVSP API functions (e.g., SetParamVal for Span, Chord, Cap Type, and Tessellation Control).
 - Output must strictly respect fluid dynamics principles (vorticity distribution, pressure gradient delta-Cp, and lift-induced drag).
 - Do not invent non-physical aerodynamic behaviors.
- The script and formulas must be production-ready and executable.

[OUTPUT FORMAT]

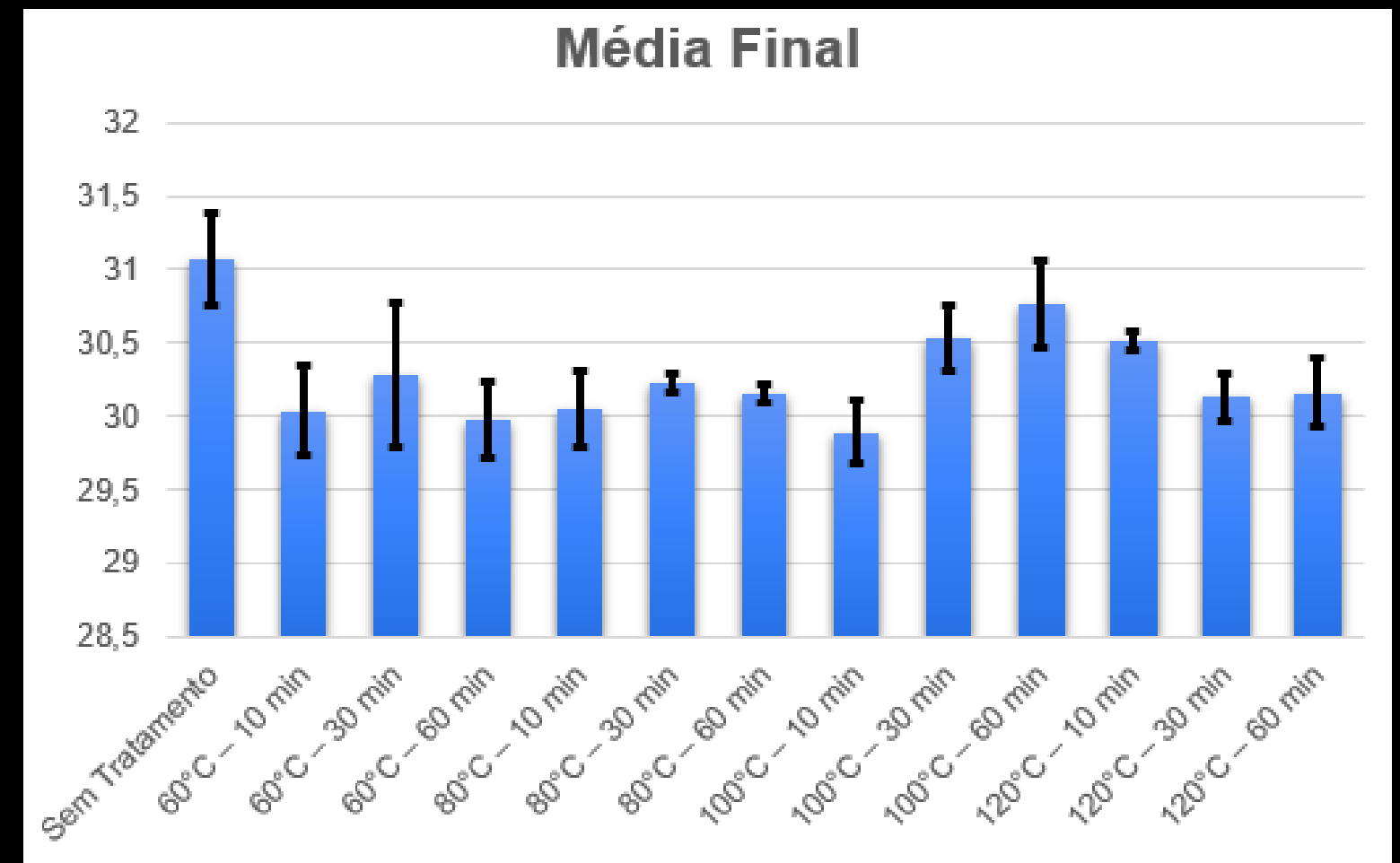
1. Complete, well-commented Python Code block for OpenVSP API execution.
2. Step-by-step mathematical validation protocol (Comparing VLM vs Panel/XFLR5 results).
3. Brief engineering insight on aerodynamic trade-offs (Lift vs. Drag penalty) based on the boundary layer behavior.

Thermal Smoothing Viability & Mechanical Integrity



Solving Problems for Tomorrow

Experimental investigation into thermal annealing as a post-processing technique to reduce surface roughness (R_a) in PLA parts printed via FDM. Analyzed 36 roughness specimens and 6 tensile test samples across combinations of temperature (60°C to 120°C) and time (10 to 60 min) following ISO 4288 and ISO 527-2 standards.



[ROLE / SYSTEM]

You are a Materials Engineer and Additive Manufacturing Specialist expert in polymer thermodynamics, Fused Deposition Modeling (FDM), and mechanical testing (ISO 527-2 / ASTM D638).

[CONTEXT]

I conducted an experimental study evaluating thermal smoothing (annealing) on FDM 3D printed Polylactic Acid (PLA) parts (glass transition $T_g \sim 55^\circ\text{C}$ - 65°C).

Experimental Dataset:

- 36 specimens for surface roughness (Ra) measured with a Mitutoyo SJ-210 profilometer (Cut-off $L_c = 2.5\text{ mm}$, $L_t = 12.5\text{ mm}$).
- Treatment matrix: 60°C , 80°C , 100°C , and 120°C for 10, 30, and 60 minutes.
- Tensile testing on EMIC DL-10000 (100 kN capacity, speed = 10 mm/min) for baseline vs. optimal condition (80°C / 30 min).
- Results: Baseline Ra = $31.07\text{ }\mu\text{m}$; $80^\circ\text{C}/30\text{ min}$ Ra = $30.23\text{ }\mu\text{m}$ (4.59% Ra improvement) UTS dropped by 4.1% (45.22 MPa down to 43.36 MPa); 120°C caused severe geometric warping and dimensional collapse.

[TASK]

1. Write a Python script using pandas and matplotlib/seaborn to calculate mean, sample standard deviation (s), and delta-Ra, and plot bar charts with error bars comparing pre- and post-thermal treatment.
2. Formulate a technical summary explaining the thermodynamic trade-off between chain relaxation/crystallization at T_g and inter-layer adhesion loss resulting in UTS reduction.

[CONSTRAINTS]

Output must strictly adhere to polymer physics (viscoelasticity, chain relaxation above T_g) and ISO 4288 metrology rules.

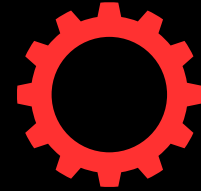
Do not hallucinate mechanical property improvements where physical degradation occurs.



AI & Prompting

Future Now

Real photos of wrinkled, deformed specimens and tensile fracture surfaces demonstrate the physical limits of polymers under thermal stress —data that synthetic AI visual models completely overlook



Thank you

Build, Create, Inspire

